Project title: Content and user preference based music recommendation system using GNN and KG **Group:** 7 — Pedro Fanica, Aarthi Perumpillichira, Maja Skwaroń

Overall assessment

The current status is promising but needs focus. The group has established a credible data pipeline and a useful collaborative-filtering baseline. The main remaining risk is that the proposed knowledge-graph/GNN model is still mostly at the design stage. Before Milestone 2, the project should move from graph design to one concrete trainable graph recommender, with a precise evaluation protocol and error analysis.

What is already working

- The project has a clear application: user-specific music recommendation.
- The datasets are well identified: Lakh MIDI Dataset, Million Song Dataset, and Echo Nest Taste Profile.
- The data integration has progressed substantially, including LMD–MSD matching and filtering.
- The group reports a filtered dataset with 286,889 users and 7,050 songs.
- A real baseline has been implemented: user-based KNN collaborative filtering.
- Preliminary baseline results are reported using Recall@10, NDCG@10, Hit Rate, Coverage, Popularity Bias, and Novelty.
- The methodology figure helps communicate the intended graph structure.

Main issues to address

- The final prediction task should be stated more precisely: top-N recommendation from implicit feedback, not generic "preference prediction".
- The knowledge graph is not yet converted into a trainable graph representation.
- The GNN model is not yet specified in enough detail.
- The evaluation protocol needs tighter definition, especially candidate sampling, negative items, and metric computation.
- There are no training curves yet, which is understandable for KNN, but Milestone 2 should include learning curves for the neural model.
- There is no qualitative error analysis yet.
- The feature space may become too large if too many symbolic music features are represented as graph nodes.

Detailed comments

1. Problem and evaluation

The problem is interesting and now much clearer than a generic music recommendation task. However, the report should explicitly define the task as: given a user's observed listening history, rank unseen songs and evaluate whether held-out listened songs appear in the top-k recommendations. This also clarifies that the target is implicit feedback, not explicit ratings.

The metrics are appropriate for top-N recommendation, especially Recall@10, NDCG@10, Hit Rate, and Coverage. Still, the report should define each metric and explain how candidate songs are selected during evaluation. For example, are models ranking all 7,050 songs, or a sampled subset of negatives plus positives? This choice strongly affects the meaning of the numbers.

2. Data pipeline

The data pipeline is one of the strongest parts of the report. The group explains the LMD–MSD alignment, DTW thresholding, filtering, user interaction processing, and removal of users with too few interactions. This is good progress.

The next version should make the pipeline reproducible: exact scripts, file sources, thresholds, filtering order, final counts after each step, and the final feature table/graph format. The feature extraction step is still described as future work. By Milestone 2, the group should freeze a first usable feature set rather than continue exploring many possible symbolic features.

3. Evaluation validity

The per-user 80/20 split is reasonable for a warm-start recommendation setting. The group should state explicitly that this project does not evaluate user cold-start, because all test users also appear in training.

There is some risk of evaluation ambiguity. Song metadata and audio/MIDI features can be available for all songs, including test songs, but interaction-derived quantities must be computed only from the training interactions. The report should also verify that recommendations exclude songs already seen by the user in training.

One point to check: the reported composite score and Recall@10 are both 0.4559. Since the composite score is a weighted combination of several metrics, this coincidence should be verified. Also, Novelty = 100% may be trivial if the recommender is explicitly forbidden from recommending training songs; in that case it should not be treated as strong evidence of model quality.

4. Baseline and preliminary results

The KNN baseline is useful and appropriate. The report should add implementation details: similarity metric, normalization of play counts, whether play counts are binarized or log-scaled, how missing values are treated, and whether the KNN is user-user or item-item in the final implementation.

The baseline results are encouraging, but they need interpretation. A table with all metrics for several k values would make the analysis clearer than reporting only the best k. The group should also compare against a very simple popularity baseline. Without this, it is hard to know whether KNN is genuinely personalizing or mostly recommending globally popular tracks.

5. Error analysis and next experiments

The report does not yet inspect successes and failures. This should be added before Milestone 2. For example, select a few users and show: their training songs, held-out songs, top-10 recommendations from KNN, and whether the recommendations are similar by artist, genre, tempo, or popularity. This will reveal whether the model is learning musical similarity, popularity, or user-neighbour effects.

The next experiments should be narrower. Instead of trying many graph designs, choose one first graph recommender. A good minimum path would be: popularity baseline → KNN collaborative filtering → LightGCN or GraphSAGE-style user-song graph model → graph model with song metadata/MIDI features.

6. Scope and feasibility

The project is feasible, but only if the graph model is simplified. Represent categorical entities such as users, songs, artists, and genres as nodes. Represent continuous musical descriptors such as tempo, loudness, pitch statistics, note density, and rhythm descriptors as song node features, not necessarily as separate nodes. This will reduce graph complexity and make the GNN easier to train.

The key scientific question should be: **do musical and metadata features improve recommendation over interaction-only collaborative filtering?** Everything else should support that comparison.

Minimum viable final project

A successful minimum version of this project would be:

- A reproducible LMD–MSD–Taste Profile pipeline with fixed filtering and train/validation/test splits.
- At least two baselines: popularity and KNN collaborative filtering.
- One implemented graph-based recommender trained end-to-end and compared against the baselines using the same evaluation protocol.
- A small ablation showing whether metadata and/or MIDI-derived features improve the graph model.
- Qualitative error analysis for representative users.

Optional extensions, only if the main experiments are stable:

- Cold-start or low-interaction user analysis.
- More detailed symbolic music feature engineering.
- More complex knowledge graph relations or multi-relational GNNs.
- Diversity and novelty analysis beyond standard accuracy metrics.

Most important changes before Milestone 2

- Define the recommendation task and evaluation protocol precisely, including candidate songs, negatives, and metric computation.
- Implement one trainable graph recommender; do not spend too much time only refining the ontology.
- Add a popularity baseline and verify the reported metric calculations.
- Freeze a compact first feature set for metadata and MIDI features.
- Add qualitative examples of recommendations and failure cases.

Grade

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